MSG No. 194.272 P_c6_3/mmc [Type IV, hexagonal]

Table 1: Wyckoff site: 2a, site symmetry: 6'/mmm'

No.	position	mapping
1	$\left[0,\ 0,\ \frac{3}{4}\right]$	[1,3,5,10,11,12,14,16,18,19,20,21,26,28,30,31,32,33,37,39,41,46,47,48]
2	$\left[0,\ 0,\ \frac{1}{4}\right]$	[2,4,6,7,8,9,13,15,17,22,23,24,25,27,29,34,35,36,38,40,42,43,44,45]

Table 2: Wyckoff site: 2b, site symmetry: 6'/m'mm'

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,3,5,7,8,9,13,15,17,19,20,21,26,28,30,34,35,36,38,40,42,46,47,48]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[2,4,6,10,11,12,14,16,18,22,23,24,25,27,29,31,32,33,37,39,41,43,44,45]

Table 3: Wyckoff site: 4c, site symmetry: -6m2

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[1,3,5,10,11,12,14,16,18,19,20,21]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	[2,4,6,7,8,9,13,15,17,22,23,24]
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[25,27,29,34,35,36,38,40,42,43,44,45]
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	[26,28,30,31,32,33,37,39,41,46,47,48]

Table 4: Wyckoff site: 4d, site symmetry: -6'm2'

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	[1,3,5,19,20,21,34,35,36,38,40,42]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[2,4,6,22,23,24,31,32,33,37,39,41]
3	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[7,8,9,13,15,17,26,28,30,46,47,48]
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	[10,11,12,14,16,18,25,27,29,43,44,45]

Table 5: Wyckoff site: 4e, site symmetry: 6'mm'

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,3,5,19,20,21,26,28,30,46,47,48]
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[2,4,6,22,23,24,25,27,29,43,44,45]
3	$[0,\ 0,\ -z]$	[7,8,9,13,15,17,34,35,36,38,40,42]
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[10,11,12,14,16,18,31,32,33,37,39,41]

Table 6: Wyckoff site: 6f, site symmetry: mmm'

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[1,11,16,19,28,31,37,47]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[2,9,17,22,29,34,38,45]
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	[3,12,18,20,30,32,39,48]

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[4,7,13,23,25,35,40,43]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[5,10,14,21,26,33,41,46]
6	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[6,8,15,24,27,36,42,44]

Table 7: Wyckoff site: 6g, site symmetry: m'mm'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,7,13,19,28,35,40,47]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,10,14,22,29,33,41,45]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,8,15,20,30,36,42,48]
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[4,11,16,23,25,31,37,43]
5	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[5,9,17,21,26,34,38,46]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,12,18,24,27,32,39,44]

Table 8: Wyckoff site: 8h, site symmetry: 3m.

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	[2,4,6,22,23,24]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9,13,15,17]
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	[10,11,12,14,16,18]
5	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	[25,27,29,43,44,45]
6	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[26,28,30,46,47,48]
7	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	[31,32,33,37,39,41]
8	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[34,35,36,38,40,42]

Table 9: Wyckoff site: 12i, site symmetry: 2'mm'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,19,28,47]
2	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[2,22,29,45]
3	$\left[0, \frac{1}{2}, z\right]$	[3,20,30,48]
4	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[4,23,25,43]
5	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[5,21,26,46]
6	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[6,24,27,44]
7	$\left[\frac{1}{2}, 0, -z\right]$	[7,13,35,40]
8	$\left[0, \frac{1}{2}, -z\right]$	[8,15,36,42]
9	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,17,34,38]
10	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[10,14,33,41]
11	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[11,16,31,37]
12	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[12,18,32,39]

Table 10: Wyckoff site: `12j`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[1,16,31,47]`
2	$\left[x,\ x,\ \frac{1}{4}\right]$	`[2,17,34,45]`
3	$\left[0,\ x,\ \frac{3}{4}\right]$	`[3,18,32,48]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,13,35,43]`
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[5,14,33,46]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,15,36,44]`
7	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[7,23,25,40]`
8	$\left[0,\ x,\ \frac{1}{4}\right]$	`[8,24,27,42]`
9	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[9,22,29,38]`
10	$\left[x,\ x,\ \frac{3}{4}\right]$	`[10,21,26,41]`
11	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[11,19,28,37]`
12	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[12,20,30,39]`

Table 11: Wyckoff site: `12k`, site symmetry: `m'2m'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7,40,47]`
2	$\left[x,\ x,\ \frac{1}{2}\right]$	`[2,10,41,45]`
3	$[0,\ x,\ 0]$	`[3,8,42,48]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,11,37,43]`
5	$[-x,\ -x,\ 0]$	`[5,9,38,46]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,12,39,44]`
7	$[-x,\ 0,\ 0]$	`[13,19,28,35]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[14,22,29,33]`
9	$[0,\ -x,\ 0]$	`[15,20,30,36]`
10	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[16,23,25,31]`
11	$[x,\ x,\ 0]$	`[17,21,26,34]`
12	$\left[0,\ x,\ \frac{1}{2}\right]$	`[18,24,27,32]`

Table 12: Wyckoff site: `12l`, site symmetry: `mm2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[1,11,16,19]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,9,17,22]`
3	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[3,12,18,20]`
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[4,7,13,23]`
5	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[5,10,14,21]`
6	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[6,8,15,24]`
7	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[25,35,40,43]`
8	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[26,33,41,46]`
9	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[27,36,42,44]`
10	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[28,31,37,47]`
11	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[29,34,38,45]`
12	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[30,32,39,48]`

Table 13: Wyckoff site: `12m`, site symmetry: `m'm2'`

No.	position	mapping
1	$[x,\ 2x,\ 0]$	`[1,19,35,40]`
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[2,22,33,41]`
3	$[-2x,\ -x,\ 0]$	`[3,20,36,42]`
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,23,31,37]`
5	$[x,\ -x,\ 0]$	`[5,21,34,38]`
6	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[6,24,32,39]`
7	$[-x,\ -2x,\ 0]$	`[7,13,28,47]`
8	$[2x,\ x,\ 0]$	`[8,15,30,48]`
9	$[-x,\ x,\ 0]$	`[9,17,26,46]`
10	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[10,14,29,45]`
11	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[11,16,25,43]`
12	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[12,18,27,44]`

Table 14: Wyckoff site: `24n`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,47]`
2	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[2,45]`
3	$[0,\ x,\ z]$	`[3,48]`
4	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[4,43]`
5	$[-x,\ -x,\ z]$	`[5,46]`
6	$\left[0,\ -x,\ z+\frac{1}{2}\right]$	`[6,44]`
7	$[x,\ 0,\ -z]$	`[7,40]`
8	$[0,\ x,\ -z]$	`[8,42]`
9	$[-x,\ -x,\ -z]$	`[9,38]`
10	$\left[x,\ x,\ \frac{1}{2}-z\right]$	`[10,41]`
11	$\left[-x,\ 0,\ \frac{1}{2}-z\right]$	`[11,37]`
12	$\left[0,\ -x,\ \frac{1}{2}-z\right]$	`[12,39]`
13	$[-x,\ 0,\ -z]$	`[13,35]`
14	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	`[14,33]`
15	$[0,\ -x,\ -z]$	`[15,36]`
16	$\left[x,\ 0,\ \frac{1}{2}-z\right]$	`[16,31]`
17	$[x,\ x,\ -z]$	`[17,34]`
18	$\left[0,\ x,\ \frac{1}{2}-z\right]$	`[18,32]`
19	$[-x,\ 0,\ z]$	`[19,28]`
20	$[0,\ -x,\ z]$	`[20,30]`
21	$[x,\ x,\ z]$	`[21,26]`
22	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	`[22,29]`
23	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[23,25]`
24	$\left[0,\ x,\ z+\frac{1}{2}\right]$	`[24,27]`

Table 15: Wyckoff site: `24o`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 2x,\ z]$	`[1,19]`
2	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[2,22]`
3	$[-2x,\ -x,\ z]$	`[3,20]`
4	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	`[4,23]`
5	$[x,\ -x,\ z]$	`[5,21]`
6	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	`[6,24]`
7	$[-x,\ -2x,\ -z]$	`[7,13]`
8	$[2x,\ x,\ -z]$	`[8,15]`
9	$[-x,\ x,\ -z]$	`[9,17]`
10	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	`[10,14]`
11	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	`[11,16]`
12	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	`[12,18]`
13	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	`[25,43]`
14	$[-x,\ x,\ z]$	`[26,46]`
15	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	`[27,44]`
16	$[-x,\ -2x,\ z]$	`[28,47]`
17	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[29,45]`
18	$[2x,\ x,\ z]$	`[30,48]`
19	$\left[-x,\ -2x,\ \frac{1}{2}-z\right]$	`[31,37]`
20	$\left[2x,\ x,\ \frac{1}{2}-z\right]$	`[32,39]`
21	$\left[-x,\ x,\ \frac{1}{2}-z\right]$	`[33,41]`
22	$[x,\ -x,\ -z]$	`[34,38]`
23	$[x,\ 2x,\ -z]$	`[35,40]`
24	$[-2x,\ -x,\ -z]$	`[36,42]`

Table 16: Wyckoff site: `24p`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{3}{4}\right]$	`[1,16]`
2	$\left[x-y,\ x,\ \frac{1}{4}\right]$	`[2,17]`
3	$\left[-y,\ x-y,\ \frac{3}{4}\right]$	`[3,18]`
4	$\left[-x,\ -y,\ \frac{1}{4}\right]$	`[4,13]`
5	$\left[-x+y,\ -x,\ \frac{3}{4}\right]$	`[5,14]`
6	$\left[y,\ -x+y,\ \frac{1}{4}\right]$	`[6,15]`
7	$\left[x-y,\ -y,\ \frac{1}{4}\right]$	`[7,23]`
8	$\left[y,\ x,\ \frac{1}{4}\right]$	`[8,24]`
9	$\left[-x,\ -x+y,\ \frac{1}{4}\right]$	`[9,22]`
10	$\left[x,\ x-y,\ \frac{3}{4}\right]$	`[10,21]`
11	$\left[-x+y,\ y,\ \frac{3}{4}\right]$	`[11,19]`
12	$\left[-y,\ -x,\ \frac{3}{4}\right]$	`[12,20]`
13	$\left[x,\ y,\ \frac{1}{4}\right]$	`[25,40]`
14	$\left[x-y,\ x,\ \frac{3}{4}\right]$	`[26,41]`
15	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	`[27,42]`
16	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[28,37]`
17	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	`[29,38]`

continued ...

Table 16

No.	position	mapping
18	$\left[y,\ -x+y,\ \frac{3}{4}\right]$	[30,39]
19	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	[31,47]
20	$\left[y,\ x,\ \frac{3}{4}\right]$	[32,48]
21	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[33,46]
22	$\left[x,\ x-y,\ \frac{1}{4}\right]$	[34,45]
23	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	[35,43]
24	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[36,44]

Table 17: Wyckoff site: 24q, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,40]
2	$\left[x-y,\ x,\ \frac{1}{2}\right]$	[2,41]
3	$[-y,\ x-y,\ 0]$	[3,42]
4	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[4,37]
5	$[-x+y,\ -x,\ 0]$	[5,38]
6	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	[6,39]
7	$[x-y,\ -y,\ 0]$	[7,47]
8	$[y,\ x,\ 0]$	[8,48]
9	$[-x,\ -x+y,\ 0]$	[9,46]
10	$\left[x,\ x-y,\ \frac{1}{2}\right]$	[10,45]
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	[11,43]
12	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[12,44]
13	$[-x,\ -y,\ 0]$	[13,28]
14	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[14,29]
15	$[y,\ -x+y,\ 0]$	[15,30]
16	$\left[x,\ y,\ \frac{1}{2}\right]$	[16,25]
17	$[x-y,\ x,\ 0]$	[17,26]
18	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[18,27]
19	$[-x+y,\ y,\ 0]$	[19,35]
20	$[-y,\ -x,\ 0]$	[20,36]
21	$[x,\ x-y,\ 0]$	[21,34]
22	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	[22,33]
23	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	[23,31]
24	$\left[y,\ x,\ \frac{1}{2}\right]$	[24,32]

Table 18: Wyckoff site: 48r, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$[-y,\ x-y,\ z]$	[3]
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[4]
5	$[-x+y,\ -x,\ z]$	[5]

continued ...

Table 18

No.	position	mapping
6	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	[6]
7	$[x-y,\, -y,\, -z]$	[7]
8	$[y,\, x,\, -z]$	[8]
9	$[-x,\, -x+y,\, -z]$	[9]
10	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	[14]
15	$[y,\, -x+y,\, -z]$	[15]
16	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[16]
17	$[x-y,\, x,\, -z]$	[17]
18	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	[18]
19	$[-x+y,\, y,\, z]$	[19]
20	$[-y,\, -x,\, z]$	[20]
21	$[x,\, x-y,\, z]$	[21]
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[24]
25	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[25]
26	$[x-y,\, x,\, z]$	[26]
27	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[27]
28	$[-x,\, -y,\, z]$	[28]
29	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[29]
30	$[y,\, -x+y,\, z]$	[30]
31	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	[31]
32	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[32]
33	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[33]
34	$[x,\, x-y,\, -z]$	[34]
35	$[-x+y,\, y,\, -z]$	[35]
36	$[-y,\, -x,\, -z]$	[36]
37	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[37]
38	$[-x+y,\, -x,\, -z]$	[38]
39	$\left[y,\, -x+y,\, \frac{1}{2}-z\right]$	[39]
40	$[x,\, y,\, -z]$	[40]
41	$\left[x-y,\, x,\, \frac{1}{2}-z\right]$	[41]
42	$[-y,\, x-y,\, -z]$	[42]
43	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	[43]
44	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[44]
45	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	[45]
46	$[-x,\, -x+y,\, z]$	[46]
47	$[x-y,\, -y,\, z]$	[47]
48	$[y,\, x,\, z]$	[48]